REVIEW

Low carbohydrate availability impairs hypertrophy and anaerobic performance

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Purpose of review
Highlight contemporary evidence examining the effects of carbohydrate restriction on the intracellular regulation of muscle mass and anaerobic performance.

Recent findings
Low carbohydrate diets increase fat oxidation and decrease fat mass. Emerging evidence suggests that dietary carbohydrate restriction increases protein oxidation, thereby limiting essential amino acid availability necessary to stimulate optimal muscle protein synthesis and promote muscle recovery. Low carbohydrate feeding for 24 h increases branched-chain amino acid (BCAA) oxidation and reduces myogenic regulator factor transcription compared to mixed-macronutrient feeding. When carbohydrate restriction is maintained for 8 to 12 weeks, the alterations in anabolic signaling, protein synthesis, and myogenesis likely contribute to limited hypertrophic responses to resistance training. The blunted hypertrophic response to resistance training when carbohydrate availability is low does not affect muscle strength, whereas persistently low muscle glycogen does impair anaerobic output during high-intensity sprint and time to exhaustion tests.

Summary
Dietary carbohydrate restriction increases BCAA oxidation and impairs muscle hypertrophy and anaerobic performance, suggesting athletes who need to perform high-intensity exercise should consider avoiding dietary strategies that restrict carbohydrate.

Keywords
glycogen, high fat, muscle anabolism, myogenesis

INTRODUCTION

Emerging evidence shows consuming a low carbohydrate diet can promote weight loss and reductions in body fat [1,2]. Decreases in fat mass following low carbohydrate intake may be due, in part, to molecular adaptations within peripheral tissue that promotes the mobilization and oxidation of fatty acids to compensate for reductions in liver and muscle glycogen [3]. These molecular adaptations rapidly (<6 days) increase lipolysis, resulting in greater rates of whole-body fat oxidation at rest and during exercise [4,5,6▪]. The potential for these diets to reduce fat mass and alter substrate oxidation has caused a resurgence of interest across athletic populations. Particularly, adhering to a low carbohydrate diet combined with an appropriate resistance training program may help athletes achieve an optimal strength and power to mass ratio by reducing total body mass, fat mass, and maintaining or potentially increasing fat-free mass [7].

While consuming a low carbohydrate diet facilitates increased fat oxidation and reductions in fat mass [7,8], these diets may also increase reliance on protein for energy yielding purposes [9,10,11▪]. Relying on dietary protein for energy production reduces the availability of essential amino acids, particularly the branched-chain amino acids (BCAA) leucine, isoleucine, and valine to maintain optimal protein synthesis rates [12,13]. Failure to support muscle anabolism following a low carbohydrate diet may limit muscle hypertrophy and compromise muscle remodeling and repair [6▪,14▪]. In this review, we will describe the current literature examining the influence of dietary carbohydrate restriction on the molecular regulation of muscle mass and

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KEY POINTS

- Low carbohydrate diets decrease body fat mass and increase whole-body and skeletal muscle fat oxidation.
- New data demonstrates that low carbohydrate diets increase the concentrations of BCAA and muscle protein breakdown metabolites, due to increased reliance on essential amino acids for oxidation.
- Increased amino acid oxidation in response to low carbohydrate availability decreases muscle remodeling and repair after exercise, likely contributing to blunted hypertrophic responses to resistance exercise training.
- Impaired muscle hypertrophy does not affect muscle strength, but persistently low glycogen following a low carbohydrate diet diminishes anaerobic performance

its subsequent effects on fat-free mass and anaerobic performance.

MUSCLE MOLECULAR ADAPTATIONS

Coupling low carbohydrate diets with exercise training diminishes muscle glycogen content [3,15]. Reduced glycogen availability modulates intramuscular signaling pathways that regulate energy metabolism by decreasing carbohydrate oxidation, while increasing both fat and protein oxidation (Fig. 1). The molecular regulation of glycolysis appears to be maintained [3,15], but low glycogen reduces the activity of pyruvate dehydrogenase (rate limiting enzyme of glycolytic flux) which limits further glycolysis in an already glycogen diminished state [3,16]. To compensate for reduced carbohydrate oxidation, transcriptional regulation of fatty acid uptake (fatty acid translocase), binding (fatty acid bind protein), transport (carnitine palmitoyl transferase, CPT), and oxidation (hydroxyacyl-CoA dehydrogenase) are increased to maintain energy production [3]. Concurrent with increases in fat oxidation, at rest and during exercise, are increased concentrations of acyl-carnitine metabolites following a low carbohydrate diet [11▪]. These mitochondria metabolites are formed when fatty acyl-CoAs are carnitized to acyl-carnitines by CPT1 to allow for the shuttling of fatty acids from the cytosol into the mitochondria signifying increased β-oxidation [11▪].

The primary modification in substrate oxidation in response low carbohydrate diets is increased fat oxidation, but there is also a notable increase in the reliance on dietary protein for oxidative and gluconeogenic purposes [9,10,11▪]. Greater protein oxidation likely contributes to lower rates of muscle protein synthesis when exercise is initiated with low glycogen content [6▪,9,10,11▪,17]. We recently reported differences in circulating metabolomics profiles that were indicative of increased BCAA catabolism when comparing participants who had low versus adequate glycogen content [11▪]. Low glycogen availability increased concentrations of urea and 3-methylhistine, and the majority of metabolites downstream of BCAA oxidation compared with adequate glycogen content [11▪]. Increased breakdown of BCAA with low glycogen suggests their carbon skeletons are used for energy production, as these amino acids may be converted to acetyl CoA or succinyl CoA and enter the tricarboxylic acid cycle (Fig. 1) [11▪]. Increased BCAA oxidation may also facilitate greater rates of β-oxidation, as acyl-carnitine metabolites isobutyrylcarnitine, butyrylcarnitine, 2-methylbutyrylcarnitine, and 3-methylglutarylcarnitine are downstream byproducts of BCAA oxidation that increase in concentration when glycogen is low [11▪].

Increased BCAA oxidation likely contributes to reductions in the transcription of myogenic regulatory factors, PAX7, MYOD, and MYOGENIN when glycogen availability is low [6▪,18]. Restricting glucose in cell culture and mouse models has shown to blunt cellular growth and reduces the expression of these myogenic regulator factors partly due to increased BCAA oxidation [18]. Low glycogen may also impair myogenesis through reduced activation of protein kinase B (Akt) [6▪]. Akt is an upstream regulator of myogenesis, governing the expression of myogenic regulator factors to stimulate myotube formation through inhibition of glycogen synthase kinase-3β (GSK-3β) [19,20]. In addition to regulating glycogen synthesis by inhibiting glycogen synthase, increased activation of GSK-3β inhibits the transcription of PAX7, MYOD, and MYOGENIN [21]. The phosphorylation status of Akt and GSK-3β are, in part, regulated by insulin [22]. Insulin-mediated phosphorylation of Akt deactivates GSK-3β, increasing the expression of myogenic regulatory factors that facilitate myogenesis [22]. Habitual (> 9 months) consumption of low carbohydrate diets reduces protein content of insulin receptor substrate (IRS) [15] and low IRS protein content may decrease Akt phosphorylation that permits GSK-3β inhibition of myogenesis [6▪]. We suspect that the reductions in myogenesis with low glycogen impairs muscle remodeling and repair after exercise.

Intramuscular availability of BCAAs stimulate the mechanistic target of rapamycin complex 1 (mTORC1), which is the central regulator of muscle protein synthesis [23,24]. Increased intramuscular concentrations of BCAAs stimulates the translocation of mTORC1 to the lysosome where it is

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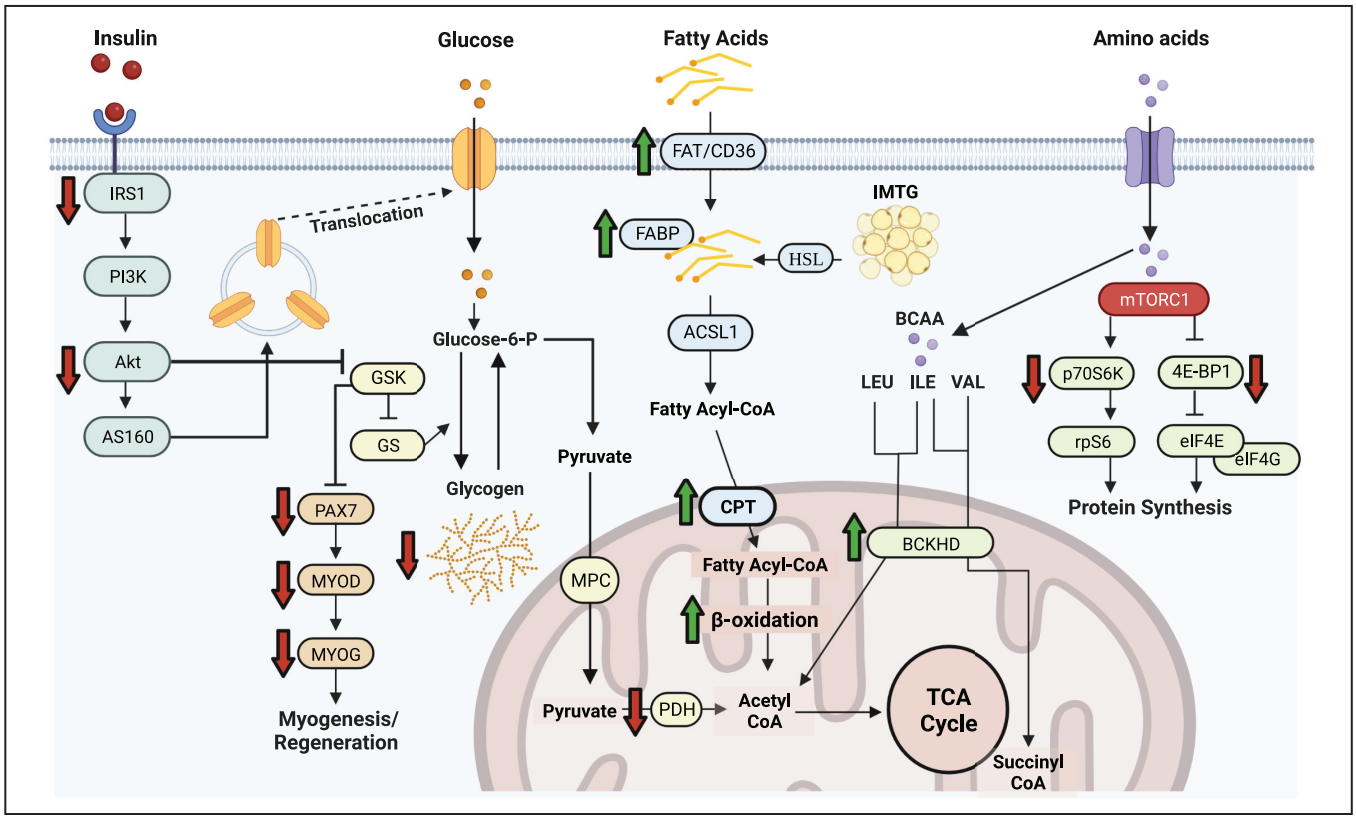

FIGURE 1. Molecular adaptations regulating carbohydrate, fat, and protein metabolism in response to low carbohydrate diets within skeletal muscle. PI3K, phosphoinositide 3-kinases; Akt, protein kinase B; AS160, Akt substrate 160; BCAA, branched-chain amino acids; BCKHD, branched-chain keto acid dehydrogenase; CPT, carnitine palmitoyl transferase; FABP, fatty acid binding protein; FAT/CD36, fatty acid translocase; GSK, glycogen synthase kinase; GS, glycogen synthase; ILE, isoleucine; IMTG, intramuscular triglycerides; IRS1, insulin receptor substrate; LEU, leucine; MPC, mitochondrial pyruvate carrier; mTORC1, mechanistic target of rapamycin complex 1; PDh, pyruvate dehydrogenase; rpS6, ribosomal protein S6; VAL, valine.

activated and increases rates of muscle protein synthesis [23,24]. Greater reliance on BCAA for energy yielding purposes with low glycogen may limit their availability to stimulate mTORC1 signaling. However, there is no consensus clearly defining the effects of glycogen availability on mTORC1 signaling, as studies report decreases or no effect on the phosphorylation status of individual proteins within the signaling cascade [6▪]. Discordant results are likely the result of differences in glycogen availability across studies. Resting glycogen concentrations for active individuals is ~450 mmol/kg/dry weight [3]. Anabolic resistance in response to exercise stimulation may not occur until muscle glycogen content is < 100 μmol/kg/dry weight [6▪]. Even with reductions in stores, if not severely restricted, sufficient glycogen availability (~200 μmol/kg/dry weight) may be adequate to maintain mTORC1 signaling [6▪]. The current understanding of what occurs when BCAAs are relied upon for fuel with low glycogen is that anabolic and myogenic signaling are reduced, which may contribute to reductions in skeletal muscle mass with chronic consumption of low carbohydrate diets.

FAT-FREE MASS

A recent systematic review by Coleman *et al.* [25] reported across 13 studies that when low carbohydrate diets are combined with exercise training, healthy, normal weight active participants have greater losses of total body and fat mass compared to participants consuming a mixed-macronutrient diet for 21 to 84 days (Fig. 2). Reductions in total body mass suggests that participants consuming a low carbohydrate diet were overall in an energy deficit. Indeed, estimating energy balance from changes in energy stores, Coleman *et al.* [25] reported a negative energy balance of 339 kcal/d. Interestingly, despite participants losing 2.9 kg of total body mass, fat-free mass was largely maintained (-0.3 kg) when consuming a low carbohydrate diet combined with exercise training [25]. Conversely, when participants consumed a

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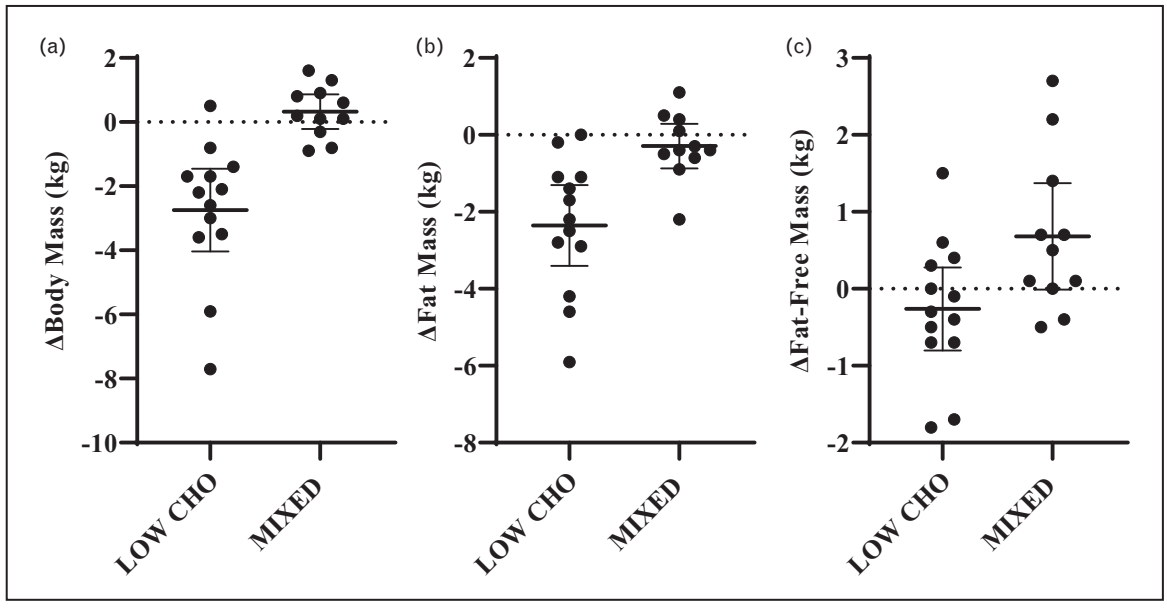

FIGURE 2. Changes in body mass (a), fat mass (b), and fat-free mass following a low carbohydrate or mixed-macronutrient diet for 21–84 days. Individual points represent a mean value from an individual study that were reported in the systematic review by Coleman *et al.* [25].

mixed-macronutrient diet in combination with exercise training, fat-free mass was increased (0.7 kg), resulting in a 1 kg difference in the change in fat-free mass compared to low carbohydrate diets [25]. While these data suggest that sustained restriction of dietary carbohydrate may blunt exercise training-induced muscle hypertrophy, changes in fat-free mass appear modest and there is a large degree of heterogeneity across the literature [25]. Modest changes in fat-free mass are likely the result of the relatively short duration (<84 days) of these previous studies [7,25]. The high degree of variability across studies may result from differences in the duration of the studies, use of controlled or free-living exercise interventions, and the severity of negative energy balance that participants were in while consuming the low carbohydrate diets [25].

The common observation of negative energy balance when consuming a low carbohydrate diet is a confounding variable that makes it difficult to delineate the mechanism resulting in discrepancies in fat-free mass responses to exercise training. Like carbohydrate restriction, negative energy balance upregulates BCAA oxidation, increasing the reliance on essential amino acid carbon skeletons for gluconeogenesis and energy production [12,13] (Fig. 3). Increased reliance on essential amino acids for energy yielding purposes results in negative protein balance [12,13], which may blunt hypertrophic responses to exercise training [26].

STRENGTH AND POWER

Despite a blunted hypertrophic response to low carbohydrate diets, there does not appear to be any negative effect on muscle strength [27]. Paoli *et al.* [14▪], recently reported body builders consuming a very low carbohydrate diet (44 g carbohydrate/d) combined with daily resistance exercise training for 8 weeks experienced no change fat-free mass, decreased circulating total testosterone and insulin-like growth factor-1 (IGF-1), but their bench press and squat 1 repetition maximum (RM) increased [14▪]. Conversely, body builders who consumed a mixed-macronutrient (488 g carbohydrate/d) diet had no change total testosterone or IGF-1, and a 2 kg increase in fat-free mass and increased 1 RM bench press and squat [14▪]. The overall null effect of low carbohydrate diets on muscle strength appears to be consistent across studies using this kind of diet intervention for 2 to 84 days [27]. These data suggest that while carbohydrate availability may alter muscle hypertrophy, it does not impact muscle strength in response to resistance exercise training.

Carbohydrate availability is an important rate limiting fuel source to maintain muscle power during continuous or repeated bouts of high-intensity exercise [16]. The point at which low carbohydrate intake negatively impacts muscle power appears to be dependent on glycogen availability. It is well established that during high-intensity exercise, glycogenolysis is a primary fuel source, as it can rapidly produce fuel in the cytoplasm when oxygen delivery

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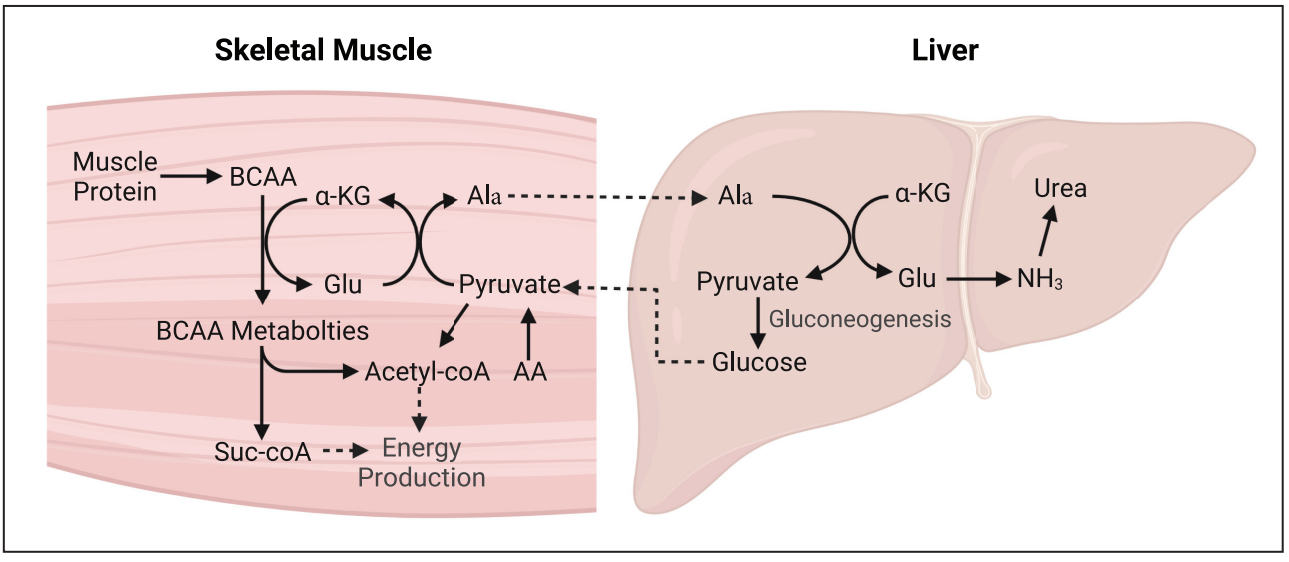

FIGURE 3. Alterations in amino acid metabolism during low carbohydrate and energy availability for oxidative and gluconeogenic purposes. Ala, alanine; BCAA, branched-chain amino acids; glu, glutamate; α-kg, α-ketoglutarate; Suc-coA, succinyl-CoA.

to the cell may be limited [28]. However, glycogen storage capacity in the muscle and liver is limited. When glycogen availability is reduced there is an associated negative impact on muscle power during high-intensity exercise [29]. As noted by Vign-Larsen *et al.* [16], there appears to be a cut point at which glycogen content impairs muscle power. Normal resting glycogen content for a trained individual is ~450 mmol/kg/dry weight [3]. Moderate reductions in muscle glycogen (350 mmol/kg/dry weight) does not affect muscle power during continuous or repeated bouts of high-intensity exercise [16]. However, when muscle glycogen is <200 mmol/kg/dry weight there is a decrease in supramaximal exercise capacity, such as time to exhaustion exercise and repeated sprint performance [16].

KNOWLEDGE GAPS OF THE CURRENT LITERATURE

The current understanding of the influence of low carbohydrate diets on fat-free mass and muscle strength and power is limited to a relatively small number of studies. Additionally, many of these published works have not controlled dietary intake. Past studies have relied primarily on instructing and educating participants on foods that contain carbohydrate [25]. This practice may be effective in facilitating lower carbohydrate intake, however not ensuring that participants adhere to a prescribed energy intake introduces potential confounding variables. Most significant is that while many studies report weight loss when participants are prescribed a low carbohydrate diet in combination with exercise training, the intent of these studies was for participants to remain in energy balance [25]. Without controlling energy intake, it is unclear if decreases in total body and fat mass are the result of metabolic adaptations or the result of participant consuming less energy due to the restrictive nature of low carbohydrate diets. Observations of negative energy balance also make it difficult to determine if observed changes in molecular signaling pathways, fat-free mass, and performance are due to low energy or carbohydrate availability. Future studies which feed participants an energy balanced, low carbohydrate diet are necessary to isolate the impact of low carbohydrate availability on molecular signaling pathways, fat-free mass, and performance.

What we know about the effects of low carbohydrate diets on protein oxidation, and myogenic and anabolic signaling pathways is limited to acute (≤24 h) diet manipulation [6▪,9,10,11▪,17]. Due to the lack of longer-term studies (>90 days), the contribution of acute molecular adaptations to changes in fat-free mass when consuming a low carbohydrate diet in combination with resistance training is unclear. Recent cross-sectional studies [15,30] assessing differences in carbohydrate and fat metabolism between endurance athletes habitually (>6 months) consuming low carbohydrate compared mixed-macronutrient diets have yielded novel information on alterations in substrate metabolism at the whole-body and skeletal muscle level. Using a similar cross-sectional design in resistance trained individuals habitually consuming a low carbohydrate diet may yield novel information that enhances what we know about the impact of altered protein oxidation, as well as myogenic and anabolic signaling on the regulation of fat-free mass.

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CONCLUSION

Low carbohydrate diets increase BCAA oxidation. Greater use of BCAA for energy production likely impairs muscle remodeling and repair after exercise, potentially blunting muscle hypertrophy. Despite lower hypertrophic responses to resistance training while consuming a low carbohydrate diet, there does not appear to be consequent impairment in muscle strength. Conversely, when low carbohydrate diets lead to reductions in glycogen stores <200 mmol/kg/dry weight there is a decrease in anaerobic capacity during high-intensity exercise. These data suggest that athletes performing high-intensity exercise should consider avoiding severe or prolonged restriction of dietary carbohydrate, because it may counter productively impair muscle hypertrophy and anaerobic performance.

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Conflicts of interest

There are no conflicts of interest.

REFERENCES AND RECOMMENDED READING

Papers of particular interest, published within the annual period of review, have been highlighted as:
■ of special interest
■■ of outstanding interest

1. Ludwig DS, Aronne LJ, Astrup A, *et al.* The carbohydrate-insulin model: a physiological perspective on the obesity pandemic. Am J Clin Nutr 2021; 114:1873–1885.
2. Hall KD, Farooqi IS, Friedman JM, *et al.* The energy balance model of obesity: beyond calories in, calories out. Am J Clin Nutr 2022; 115: 1243–1254.
3. Margolis LM, Wilson MA, Whitney CC, *et al.* Exercising with low muscle glycogen content increases fat oxidation and decreases endogenous, but not exogenous carbohydrate oxidation. Metabolism 2019; 97:1–8.
4. Burke LM, Whitfield J, Heikura IA, *et al.* Adaptation to a low carbohydrate high fat diet is rapid but impairs endurance exercise metabolism and performance despite enhanced glycogen availability. J Physiol 2021; 599:771–790.
5. Sukkar SG, Muscaritoli M. A clinical perspective of low carbohydrate ketogenic diets: a narrative review. Front Nutr 2021; 8:642628.
6. ■ Margolis LM, Wilson MA, Whitney CC, *et al.* Initiating aerobic exercise with low glycogen content reduces markers of myogenesis but not mTORC1 signaling. J Int Soc Sports Nutr 2021; 18:56.

Study findings demonstated that decreased glycogen availability following 24 h of low carbohydrate intake decreased the expression of PAX7, MYOD, and MYOGENESIS, which likely impairs muscle remodeling and repair after exercise.

7. Valenzuela PL, Castillo-Garcia A, Lucia A, Naclerio F. Effects of combining a ketogenic diet with resistance training on body composition, strength, and mechanical power in trained individuals: a narrative review. Nutrients 2021; 13:3083.
8. Murphy NE, Carrigan CT, Margolis LM. High-fat ketogenic diets and physical performance: a systematic review. Adv Nutr 2021; 12:223–233.
9. Larsen MS, Holm L, Svart MV, *et al.* Effects of protein intake prior to carbohydrate-restricted endurance exercise: a randomized crossover trial. J Int Soc Sports Nutr 2020; 17:7.
10. Gillen JB, West DWD, Williamson EP, *et al.* Low-carbohydrate training increases protein requirements of endurance athletes. Med Sci Sports Exerc 2019; 51:2294–2301.
11. ■ Margolis LM, Karl JP, Wilson MA, *et al.* Serum branched-chain amino acid metabolites increase in males when aerobic exercise is initiated with low muscle glycogen. Metabolites 2021; 11:828.

Study findings demonstated that following 24 h of low carbohydrate diets increases the concentrations of branched-chain amino acids and muscle protein breakdown metabolites, indicating increased reliance on essential amino acids for oxidation.

12. Gwin JA, Church DD, Hatch-McChesney A, *et al.* Essential amino acid-enriched whey enhances postexercise whole-body protein balance during energy deficit more than iso-nitrogenous whey or a mixed-macronutrient meal: a randomized, crossover study. J Int Soc Sports Nutr 2021; 18:4.
13. Gwin JA, Church DD, Hatch-McChesney A, *et al.* Effects of high versus standard essential amino acid intakes on whole-body protein turnover and mixed muscle protein synthesis during energy deficit: a randomized, crossover study. Clin Nutr 2021; 40:767–777.
14. ■ Paoli A, Cenci L, Pompei P, *et al.* Effects of two months of very low carbohydrate ketogenic diet on body composition, muscle strength, muscle area, and blood parameters in competitive natural body builders. Nutrients 2021; 13:374.

Study findings highlight discrepancies between muscle hypertrophy and strength, with 8 weeks of a low carbohydrate diet impair gains is muscle mass, but not gains in 1 repetition maximum for bench press and squat.

15. Webster CC, van Boom KM, Armino N, *et al.* Reduced glucose tolerance and skeletal muscle GLUT4 and IRS1 content in cyclists habituated to a long-term low-carbohydrate, high-fat diet. Int J Sport Nutr Exerc Metab 2020; 30:210–217.
16. Vigh-Larsen JF, Ortenblad N, Spriet LL, *et al.* Muscle glycogen metabolism and high-intensity exercise performance: a narrative review. Sports Med 2021; 51:1855–1874.
17. Hammond KM, Impey SG, Currell K, *et al.* Postexercise high-fat feeding suppresses p70S6K1 activity in human skeletal muscle. Med Sci Sports Exerc 2016; 48:2108–2117.
18. Shao D, Villet O, Zhang Z, *et al.* Glucose promotes cell growth by suppressing branched-chain amino acid degradation. Nat Commun 2018; 9:2935.
19. De Angelis L, Balasubramanian S, Berghella L. Akt-mediated phosphorylation controls the activity of the Y-box protein MSY3 in skeletal muscle. Skelet Muscle 2015; 5:18.
20. Liu Y, Liu P, Hu Y, *et al.* Cold-Induced RNA-binding protein promotes glucose metabolism and reduces apoptosis by increasing AKT phosphorylation in mouse skeletal muscle under acute cold exposure. Front Mol Biosci 2021; 8:685993.
21. Ragozzino E, Brancaccio M, Di Costanzo A, *et al.* 6-Bromoindirubin-3′-oxime intercepts GSK3 signaling to promote and enhance skeletal muscle differentiation affecting miR-206 expression in mice. Sci Rep 2019; 9:18091.
22. Litwiniuk A, Pijet B, Pijet-Kucicka M, *et al.* FOXO1 and GSK-3beta are main targets of insulin-mediated myogenesis in C2C12 muscle cells. PLoS One 2016; 11:e0146726.
23. Szwed A, Kim E, Jacinto E. Regulation and metabolic functions of mTORC1 and mTORC2. Physiol Rev 2021; 101:1371–1426.
24. Holowaty MNH, Lees MJ, Abou Sawan S, *et al.* Leucine ingestion promotes mTOR translocation to the periphery and enhances total and peripheral RPS6 phosphorylation in human skeletal muscle. Amino Acids 2023; 55:253–261.
25. Coleman JL, Carrigan CT, Margolis LM. Body composition changes in physically active individuals consuming ketogenic diets: a systematic review. J Int Soc Sports Nutr 2021; 18:41.
26. Paulussen KJM, McKenna CF, Beals JW, *et al.* Anabolic resistance of muscle protein turnover comes in various shapes and sizes. Front Nutr 2021; 8:615849.
27. Henselmans M, Bjornsen T, Hedderman R, Varvik FT. The effect of carbohydrate intake on strength and resistance training performance: a systematic review. Nutrients 2022; 14:856.
28. Margolis LM, Allen JT, Hatch-McChesney A, Pasiakos SM. Coingestion of carbohydrate and protein on muscle glycogen synthesis after exercise: a meta-analysis. Med Sci Sports Exerc 2021; 53:384–393.
29. Hearris MA, Hammond KM, Seaborne RA, *et al.* Graded reductions in preexercise muscle glycogen impair exercise capacity but do not augment skeletal muscle cell signaling: implications for CHO periodization. J Appl Physiol (1985) 2019; 126:1587–1597.
30. Volek JS, Freidenreich DJ, Saenz C, *et al.* Metabolic characteristics of keto-adapted ultra-endurance runners. Metabolism 2016; 65:100–110.

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